Miss Dua Agha . . $\exists x\,(Px)$

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
FSCS – Midterm Examinations Spring 2026

Time: 90 minutes **CT162 – Discrete Structures (Paper B)** **Max Marks: 20**

Note: Attempt all questions.

Q1: Determine whether the following function is injective, surjective, or a bijection from R to R. Justify your answer. **[CLO-1 2 marks]**

$f(x) = (3x + 5)/(2x - 1)$ 0 → -5 1 — 8

Q2: Prove both distributive laws of set theory for the following sets: Let A = {x | x is a multiple of 2 and less than 12}, B= {x | x is an odd positive integer less than 10 & C= {x | x is a prime number less than 15} **[CLO-1 3 marks]**

Q3: Translate the following statement into propositional logic using the propositions provided. Then, for each condition, state its converse, contrapositive, and inverse: "I will go hiking only if it is sunny and I am not busy, and if I feel hungry, I will eat or drink water"
[Use G=Go hiking, S= It is sunny, B= Busy, H= Hungry, E= Eat, D= Drink Water] **[CLO-1 3 marks]**

Q4: Prove De Morgan's Laws for Quantifiers for given statement: **[CLO-2 3 marks]**
a) There is a book in the library that is overdue. b) Every book in the library is returned on time.

Q5: Are these system specifications consistent? Use propositional variables: W: "Wi-Fi is off", C: "New connections are queued", N: "Network is functioning normally", S: "Data is sent to the server" **[CLO-2 3 marks]**
"If the Wi-Fi is not off, then new connections will be queued. If the Wi-Fi is not off, then the network is functioning normally, and conversely. If new connections are not queued, then data will be sent to the server. If the Wi-Fi is not off, then data will be sent to the server. Data will not be sent to the server."

Q6: Show that $(r \wedge s) \rightarrow (r \vee s) \equiv T$ are logically equivalent by developing a series of logical equivalences. **[CLO-3 3 marks]**

Q7: Write algorithm for bubble sort and sort [f, a, d, c, b, e] showing the lists obtained at each step. Also, mention the complexity of algorithms. **[CLO-3 3 marks]**